

Master's Comprehensive Exam

Part II

Applied (90 minutes)

December 9, 2000

Instructions: Answer three out of four questions. Open book and open notes.

- 1) A Gallup poll of 500 randomly chosen potential voters yielded the following result:

200 would vote for candidate A
150 would vote for candidate B

The rest would either abstain or vote for others. Let p_1 and p_2 be the fraction of voters that A and B have in the population. Find a large sample confidence interval for $p_1 - p_2$, at the 95% coverage level.

- 2) A report recently stated that the prices of oranges were expected to rise sharply due to low projected harvests. Here are the average prices California growers charged for a 75-pound box of navel oranges and the size of the harvests (in millions of boxes) for six consecutive years.

Harvest	72	69	58	70	65	54
Price	\$5.40	6.10	9.30	6.50	7.20	13.40

- Construct a scatterplot and find the least square estimates of β_0 and β_1 in the model $y = \beta_0 + \beta_1 x_1 + \varepsilon$.
- Test the hypothesis $\beta_1 = 0$.
- Calculate R^2 . How much of the variability in price is explained by the size of the harvest?
- Find the residuals associated with the least squares equation. Plot the residuals and comment on the results. Do you recommend revising the model.
- Using the model found in part a, find the predicted price for a box of oranges if the projected harvest is 60 million boxes; 30 million boxes. Which prediction do you think is more reliable? Why?

3. Samples of six different brands of diet/imitation margarine were analyzed to determine the level of physiologically active polyunsaturated fatty acids (PAPFUA, in percentages), resulting in the following data:

<i>Imperial:</i>	14.1,	13.6,	14.4,	14.3	
<i>Parkay:</i>	12.8,	12.5,	13.4,	13.0,	12.3
<i>Blue Bonnet:</i>	13.5,	13.4,	14.1,	14.3	
<i>Chiffon:</i>	13.2,	12.7,	12.6,	13.9	
<i>Mazola:</i>	16.8,	17.2,	16.4,	17.3,	18.0
<i>Fleischmann's</i>	18.1,	17.2,	18.7,	18.4	

- Use ANOVA to test for differences among the true average PAPFUA percentages for the different brands.
- Use Scheffé's method to compute confidence intervals for all $(\mu_i - \mu_j)$'s.
- Mazola and Fleischmann's are corn based while the others are soybean based. Compute a Scheffé interval for the contrast

$$\frac{(\mu_1 + \mu_2 + \mu_3 + \mu_4)}{4} - \frac{(\mu_5 + \mu_6)}{2}$$

4. A certain type of flashlight is sold with the four batteries included. A random sample of 150 flashlights is obtained and the number of defective batteries in each is determined, resulting in the following data:

<i>Number defective</i>	0	1	2	3	4
<i>Frequency</i>	26	51	47	16	10

Let X be the number of defective batteries in a randomly selected flashlight. Test the null hypothesis that the distribution of X is $\text{Bin}(4, \theta)$. That is, with

$p_i = P(i \text{ defectives})$, test

$$H_0: p_i = \binom{4}{i} \theta^i (1-\theta)^{4-i} \quad i = 0, 1, 2, 3, 4$$

Hint: To obtain the maximum likelihood estimate of θ , write the likelihood (the function to be maximized) as $\theta^u (1-\theta)^v$ where the exponents u and v are linear functions of the cell counts. Then take the natural log, differentiate with respect to θ , equate the result to zero, and solve for $\hat{\theta}$.